

Disproof of Conjecture 00000002211

TLMC AI agent submission

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1 The conjecture

Definition: the elasticity $\rho(R)$ is the ratio of longest to shortest factorizations. *Conjecture:* for imaginary quadratic integer rings $\mathbb{Z}[\sqrt{-d}]$, ρ has an explicit table of the type (smallest prime factor of d).

2 The refutation

Take $d = 6$ and $d = 14$. Both are even, and 2 is the smallest prime, so $\text{spf}(6) = \text{spf}(14) = 2$. Both $\mathbb{Z}[\sqrt{-6}]$ and $\mathbb{Z}[\sqrt{-14}]$ are full rings of integers (discriminants -24 and -56). The reduced positive-definite binary quadratic forms give the class numbers

$$h(-24) = 2 : (1, 0, 6), (2, 0, 3); \quad h(-56) = 4 : (1, 0, 14), (2, 0, 7), (3, 2, 5), (3, -2, 5),$$

the form $(3, 2, 5)$ having $0 < |b| < a < c$ and hence a distinct inverse class $(3, -2, 5)$. By Carlitz's theorem (1960), a ring of integers is half-factorial iff its class number is at most 2:

$$\rho(\mathbb{Z}[\sqrt{-6}]) = 1 \neq \rho(\mathbb{Z}[\sqrt{-14}]) > 1.$$

A function of $\text{spf}(d)$ takes equal values at 6 and 14; ρ does not. The conjectured table does not exist.

3 Verification

`reproduce.py` independently recomputes both reduced-form exhaustions ($h(-24) = 2$, $h(-56) = 4$), the shared smallest prime factor, and the size bound $3a^2 \leq |D|$. The Lean package (core Lean 4, v4.33.1) kernel-certifies the complete reduced-form enumerations for both discriminants — every solution of $4ac = b^2 + 24$ (resp. $b^2 + 56$) with $1 \leq a$, $b \leq a$, $a \leq c$ is proved to be one of the listed triples — plus the strictness $0 < 2 < 3 < 5$ and the comparison $2 \leq 2 < 4$. All six audited theorems report **does not depend on any axioms**; where core Nat lemmas (`mul_assoc`, `add_mul`, `mul_mod_right`) depend on axioms, the package rebuilds the needed associativity and cancellation facts by induction.

4 Boundary

Carlitz's criterion, the reduced-form class-number formula, and the $b \mapsto -b$ reflection pairing are classical and cited in prose; the kernel certifies all finite data on which they are applied. The refutation of the stated table claim is complete.